

Planning report

Fluid-thermal Analysis of the Green Revolution Energy Converter (GREC)

Mathias Ahlberg

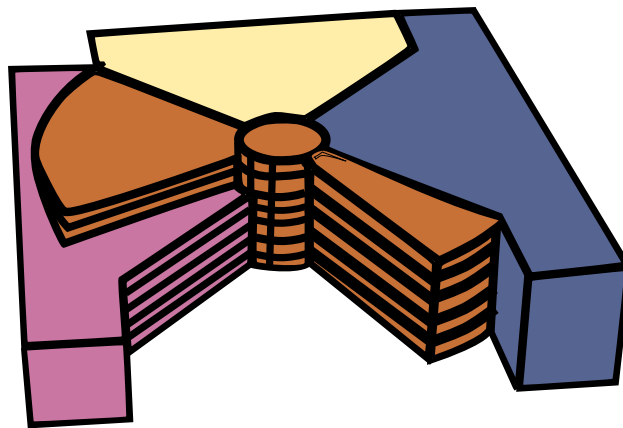
Gustav Sommerland

Carolina Fredriksson

Mukundan Subbaiya

Ankush J.Reddy

Shanmugarajan R.Ramasamy



1 Problem formulation

The Green Revolution Energy Converter (GREC) concept has progressed through theoretical modeling, internal/external heat-transfer studies, prototype development, laboratory validation and international numerical verification. This previous development includes CFD studies as well as a validation based on reproducing pressure oscillations at 3 Hz at 80°C. This verified the technology proof-of-concept from a fluid-thermal perspective. There are still many aspects in the GREC technology that needs further investigation. To understand this technology further, this project focuses on targeted experimental measurements of key operating parameters, followed by CFD verification, validation and extensive CFD parameter studies.

1.1 Objective

The overall objective of the project is to develop and validate a reliable and computationally efficient CFD methodology for the GREC LabModel v3. This should with experimental and numerical results lead to improve the understanding of the systems performance.

Experimental measurements:

- Characterize the GREC LabModel v3 experimentally by measuring pressure, temperature, rotational speed and the heat input.
- Establish repeatable steady-state operating conditions.
- Identify and reduce avoidable losses where possible.
- Estimate the remaining losses and uncertainties in the experimental setup.

CFD model:

- Develop a reliable CFD model of the GREC LabModel v3.
- Validate the CFD model against experimental measurements by using a pre-defined and justified validation approach.
- Evaluate suitable CFD modeling strategies with respect to accuracy, numerical robustness and computational cost.
- Select a suitable modeling approach for the parameter study.

Parameter study:

- Use the validated CFD methodology to investigate selected parameters affecting GREC performance.
- Select and motivate parameters from the following categories: Operating parameters, working media and geometric modifications.
- Evaluate their influence on relevant performance quantities such as pressure response, power density and efficiency of the system.
- Identify important performance and modeling limitations of the system.
- Use the results to improve understanding of GREC scaling behavior and provide guidance for future GREC development and test-rig designs.

1.2 Aim

The aim of the project is to improve the understanding of the GREC technology and the current LabModel v3. The improved understanding should serve as a solid support for the development of a next-generation prototype. The long term aim in which this project seeks to aid is the ability to turn waste/low-grade heat sources into usable electrical power.

2 Methodology Overview

2.1 CFD Methodology

2.1.1 Software

The simulations will be run in Ansys Fluent as proposed in the project proposal [1]. The first choice is Fluent's GPU solver, which might not support all features that the model needs. Early in the project a small test case will show us what works on the GPU. Anything it cannot handle will run on the CPU. Post-processing is performed in ParaView.

2.1.2 Geometry and domain

The rig has identical layers, so the model only contains one of them. The domain is cut in the middle of the fins above and below, with symmetry conditions on these planes, which requires far fewer cells than a model of the full stack. The geometry is from LabModel v3 CAD.

There will be two versions of the domain. Most simulations use a simple domain containing only the gas inside the Work Generating Volume (WGV). A full domain includes the revolving shutter, fins, insulating blocks and the small gaps between shutter and fins.

2.1.3 Solver settings

The gas moves slowly inside the rig, therefore, pressure based solver would be the ideal solver choice. The simulations will be transient as the gas temperature and pressure will change with each revolution. A steady state solution would average out the pressure oscillations, and that oscillation is what the project is supposed to predict.

The air is assumed to be an ideal gas. It is enclosed in the rig and its density must vary with temperature and pressure for any pressure oscillation to occur. Fluent's floating operating pressure option allows the absolute pressure to rise and fall in a closed domain without inlets or outlets. The energy equation will be turned on, the viscosity and conductivity of air are temperature dependent and discretization of second order will be the preferred choice.

2.1.4 Flow model

The simulations will use RANS based turbulence models, which keeps the computational cost low enough for the parameter study.[1]. SST $k-\omega$ model [2] will be used as it is better at resolving the near wall region than $k-\epsilon$, which uses wall functions. Previous CFD work found Reynolds numbers in the WGV of up to about 3000 at 3 Hz [3] so the flow at the operating speed of the rig may be near to transition. For this reason also the Transition SST model [4] is tested. If none of them reach the expected accuracy, more advanced CFD methods such as LES or DDES will be evaluated.

2.1.5 Boundary Conditions

The thermocouples on the rig give fixed temperatures for hot and cold fins. The insulating fins are walls with no heat flow, the shutter walls rotate with the gas and the mid-fin planes are symmetry planes. In the full domain, the model solves for the temperatures in the fins and block itself, and the fixed temperatures propagate to the heat block faces.

2.1.6 Mesh

A swept hexahedral mesh will be used, which will suit the thin layered geometry of the rig, with the inflation layers and the smallest cells along the fins walls where the heat transfer takes place. Identical cell sizes will be used on both sides of the sliding mesh. The mesh quality will then be evaluated based on skewness, orthogonal quality and aspect ratio and changes will be made until the mesh quality satisfies all the criteria. Then a mesh independence study will be done.

2.1.7 Time step and periodic state

The time step is a small fraction of one revolution, fixed through a time-step study. Each run starts with still gas at a mean temperature and continues until the pressure curve stops changing from one cycle to the next. Results are taken only from that repeating cycle.

2.1.8 Model Verification and validation

The model has been verified through mesh and time-step studies, each of which employed at least three levels of refinement. An energy balance is carried out on each run: when the cycle has repeated, the amount of heat absorbed at the hot fins must equal the amount of heat rejected at the cold fins, since the gas volume is fixed and therefore does no work. The simple domain is compared with the full domain to ensure that the simplifications it involves are valid.

The model will be validated against experimental data from LabV3 model at several velocities, temperature and pressure

2.1.9 Post processing

The results will be processed in Paraview and Matlab. The main quantity is pressure over one cycle since this is what the rig measurements can be compared to. The gas temperature and heat flow at the fins are to be extracted, together with contour and vector plots of the flow inside the WGV to show the pressure oscillation build up over a revolution. Once the first simulations are in progress, it will be determined which quantities are to be monitored in detail.

2.2 Experimental Methodology

All experiments need to be thought-out and motivated, meaning there needs to be a solid reason for doing the intended experiment with a meaningful intended outcome.

2.2.1 Setup the experimental model

The GREC LabModel v3 rig is instrumented to measure the major parameters, i.e temperature , pressure , rotational speed, and the heat source input. A schimatic describing where the major measurements are taken on the rig should be established in this stage. The ranges and the accuracy from the instruments while measuring the major parameters should also be documented.

2.2.2 Baseline experiments and setup assessment

The goal of this step is to establish a repeatable reference condition. This includes leak identification, confirm an even temperature distribution and rotational-speed stability. Here, it is important to create boundary conditions which can be applied in the CFD model and to identify (or eliminate) error sources that interfere with CFD validation.

2.2.3 Test matrix and operating parameters

Set up a test matrix, where the tests are described, i.e values of input parameters and also which output parameters are to be monitored. The test matrix should include the tests that needs to be done for the validation part.

2.2.4 Measurement procedure and periodic operation

Before the measurements are done, a periodic steady state criteria needs to be defined. This will be done by looking at the pressure output at stable temperatures and rotational speed. During the measurements, record the sampling rate, recording duration and test repetitions. Specify if there are any deviations from the planned procedure.

2.2.5 Heat losses and measurement uncertainty

Evaluate uncertainty sources, which includes sensor accuracy and measurement repeatability. Distinguish between heater power input from heat transferred to the gas. Upon repeated experiments, evaluation of how measurement uncertainties affect quantities (particularly pressure amplitude and temperature difference) by calculating the error factor for the output quantities.

2.2.6 Data processing

For each recorded condition, the pressure and temperature are checked for data quality and used to extract the mean pressure, peak-to-peak pressure variation, and hot/cold temperature difference. Variation between the repeated runs at each operating point is evaluated to distinguish genuine trends across the test matrix from measurement scatter.

2.2.7 Experimental data for CFD validation

Experimental and numerical conditions should be compared, distinguishing CFD inputs from independent validation measurements, meaning data used to setup the CFD model should not be used for validation. Pressure responses should be compared using predefined criteria that account for experimental uncertainty. Finally, an assessment should be made whether agreement holds across different operating points, identifying areas of model accuracy degradation.

3 Overview

The project follows the three-phase structure proposed by the supervisor. In Phase 1, reliable simulation models of the existing LabModel v3 are developed using GPU-compatible finite-volume solvers (Ansys Fluent) for a defined set of operating conditions, with different modelling strategies (e.g. mesh resolution, turbulence model) compared in terms of accuracy versus computational cost. In Phase 2, the validated methodology is used to broadly evaluate GREC performance – focusing on power density and efficiency – across the wider parameter space defined in WP3, in order to identify performance and modelling limitations and to derive scaling laws and optimisation guidelines for the next-generation rig.

If time allows, Phase 3 will explore metamodels capable of predicting performance without requiring a full CFD run for every design point. Work will begin with an experimental measurement campaign on LabModel v3 (WP1), evaluating the impact of pressure, temperature, and rotational speed on the pressure oscillations under normal operating conditions (geometric parameters are not varied experimentally, given the fixed rig). These measurements provide the reference data set for CFD verification and validation (WP2). CFD development follows the approach demonstrated in earlier GREC CFD work (ANSYS/COMSOL studies at LiU and ICAM, and the OpenFOAM chtMultiRegionFoam replication effort): a rotating Work Generating Volume enclosed by conducting hot/cold fins, with moving-mesh or MRF-based rotation and conjugate heat transfer between the fins and the WGV air. The initial focus is on RANS-based methods for computational efficiency, enabling a larger design-of-experiments in WP3; more sophisticated turbulence/CFD strategies (e.g. transient/LES approaches) will be considered only if RANS validation does not meet expectations.

The WP3 parameter study builds on the validated WP2 methodology and the WP1 measurement/sensitivity data to define the boundaries of the design space, covering operating parameters, working media, and geometric modifications, as well as start-up and degraded-operation scenarios. The number of design points will be scaled to the available computational budget, using LiU’s NSC (National Supercomputer Centre) HPC resources where appropriate. Results will be interpreted with reference to the analytical MATLAB thermodynamic model (p - V cycle, first law and ideal-gas relations) used throughout the GREC project series, allowing power density (Φ) and efficiency (η) to be compared directly against the ICAM reference case and against competing technologies (e.g. Ericsson-cycle KEOS modules, ORC/SRC Rankine systems).

In organisational terms, the project is structured around the three work packages introduced above – experimental characterisation of the LabModel v3 (WP1), CFD model development and verification/validation against that experimental data (WP2), and a CFD-based parameter study of operating conditions, working media, and geometric modifications (WP3) – which run partly in parallel, since the experimental campaign both informs and validates the CFD work, as illustrated in the Gantt chart (Sec. 4).

4 Gantt chart

The 18-week plan below distributes the work across the WP1-WP3 structure described above, together with the mandatory course milestones (planning report, mid-way report and presentation, preliminary final report, and final report/presentation). The optional Phase 3 metamodeling task is scheduled last and will only be pursued if time remains after WP3 is complete.

Project plan

Fluid-Thermal Analysis of the Green Revolution Energy Converter (GREC)

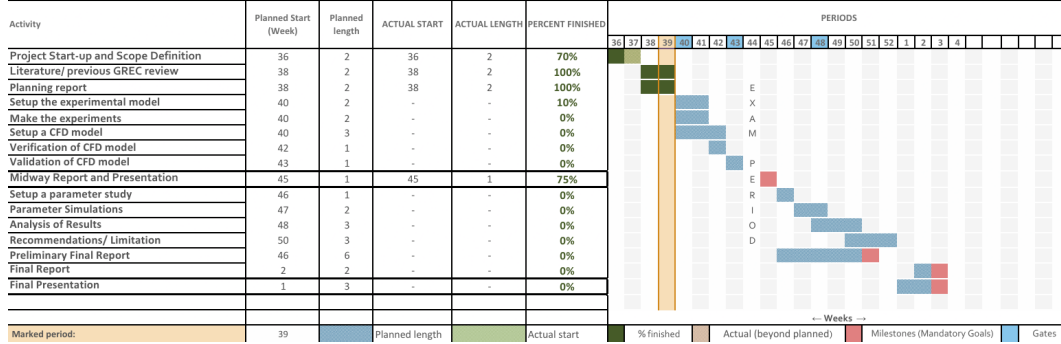


Figure 1: Preliminary Gantt chart for the GREC project (TMPM10).

Progress is tracked through monthly Project Progression Reports (PPR) and the course's mandatory milestones: the planning report, a midway report and presentation, a preliminary final report, and the final report and presentation. The responsibility for organizing documentation, submitting the PPR and communicating with the examiner rotates between the six members of the group in six roughly equal periods throughout the project, as shown in Table 1.

Period	Project Leader	Weeks
1	Mukundan Subbaiya	39–41
2	Mathias Ahlberg	42–44
3	Carolina Fredriksson	45–47
4	Shanmugarajan R. Ramasamy	48–50
5	Ankush J. Reddy	51–1
6	Gustav Sommerland	2–4

Table 1: Project leader rotation for the GREC project.

5 Appendix

The project suggested work packages:

- WP1: Experimental measurements. Conduct a series of measurements on the existing LabModel v3 under normal operating conditions, i.e. when the rig reaches a steady state behavior for the pressure oscillations. Evaluate the impact of, for example, pressure, temperature, rotational speed, and heat input. Geometric parameters will not be analyzed.
- WP2: CFD analysis - Verification & Validation. Develop reliable CFD models of the existing LabModel v3 for multiple design points under normal operating conditions. The methods will be validated using measurement data from WP1. The CFD models and simulations will be developed in Ansys Fluent or StarCCM+, enabling the possibility to use the GPU-solvers. The initial focus will be on RANS-based methods, which are less computationally intensive, thereby facilitating larger parameter studies in WP3. If this model validation does not meet expectations, more sophisticated CFD modeling strategies should be evaluated.
- WP3: CFD analysis - Parameter studies. Conduct an extensive parameter study, building upon the measurement series from WP1 using the validated CFD methodology developed in WP2. The number of design points in this study will be determined by the computational cost in relation to available computing resources. Data from the measurement series and sensitivity analyses will also be used to define the boundaries of the design space. The analysis will, for example, cover (but not excluded to): Operating parameters (pressure, temperature, rotational speed, and heat influx), Working media (air, helium, CO, refrigerant), Geometric modifications (WGV radius, layer thickness), Normal operating conditions as well as various contingency scenarios (startup transient, varying temperature, start/stop cycles, degraded cooling)

Activity	Planned Start (Week)	Planned length	ACTUAL START	ACTUAL LENGTH	PERCENT FINISHED
Project Start-up and Scope Definition	36	2	36	2	70%
Literature/ previous GREC review	38	2	38	2	100%
Planning report	38	2	38	2	100%
Setup the experimental model	40	2	-	-	10%
Make the experiments	40	2	-	-	0%
Setup a CFD model	40	3	-	-	0%
Verification of CFD model	42	1	-	-	0%
Validation of CFD model	43	1	-	-	0%
Midway Report and Presentation	45	1	45	1	75%
Setup a parameter study	46	1	-	-	0%
Parameter Simulations	47	2	-	-	0%
Analysis of Results	48	3	-	-	0%
Recommendations/ Limitation	50	3	-	-	0%
Preliminary Final Report	46	6	-	-	0%
Final Report	2	2	-	-	0%
Final Presentation	1	3	-	-	0%
Marked period:	39		Planned length		Actual start

Figure 2: Activity and weeks for preliminary Gantt chart for the GREC project (TMPM10).

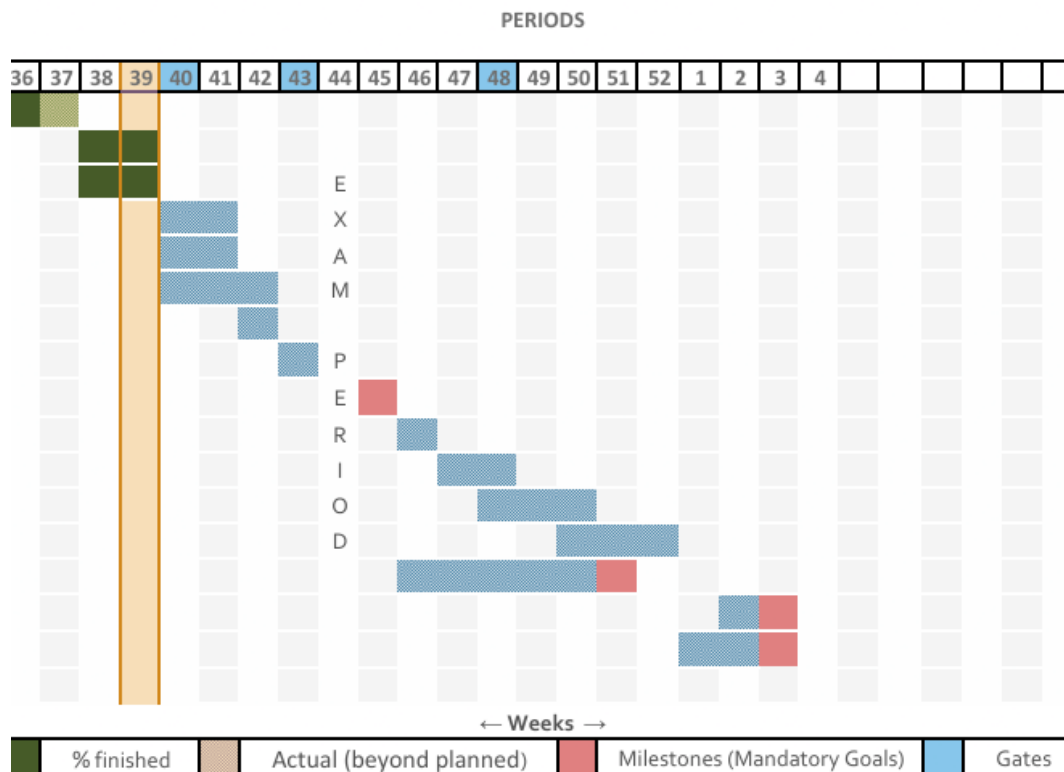


Figure 3: Visualisation overview of preliminary Gantt chart for the GREC project (TMPM10).

6 Author Contributions and Declarations

6.1 Contribution statement

- Mathias Ahlberg - MA
- Gustav Sommerland - GS
- Carolina Fredriksson - CF
- Mukundan Subbaiya - MS
- Ankush J Reddy - AJR
- Shanmugarajan R R - SRR

The authors confirm contribution to the report as follows: Problem Formulation: CF, MA, GS. Methodology Overview: AJR, MA, CF, GS. Overview: MS. Gantt Chart: CF,AJR, MA, MS, GS. Draft report preparation:AJR, CF, MA, SRR, MS, GS. All authors reviewed the results, approved the final version of the thesis report and declare that each contribution to the work from other peoples published works or unpublished sources have been acknowledged and source of information have been referenced.

6.2 Use of generative AI and AI-assisted technologies

During the preparation of this work the author(s) used Claude AI / ChatGPT in order to Academic writing and Brainstorming Ideas. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the planning report content.

References

- [1] M. Andersson, “Fluid-thermal analysis of the Green Revolution Energy Converter (GREC).” Project proposal, TPM10, Division of Applied Thermodynamics and Fluid Mechanics, Linköping University, 2026. Dated 3 September 2026.
- [2] F. R. Menter, “Two-equation eddy-viscosity turbulence models for engineering applications,” *AIAA Journal*, vol. 32, no. 8, pp. 1598–1605, 1994.
- [3] M. Peyrony-Rapatout and P. Le Provost, “Final GREC presentation.” Project presentation, ICAM Toulouse, 2025. Presented 14 January 2025.
- [4] R. B. Langtry and F. R. Menter, “Correlation-based transition modeling for unstructured parallelized computational fluid dynamics codes,” *AIAA Journal*, vol. 47, no. 12, pp. 2894–2906, 2009.